EE3072 Dynamic Systems and Control

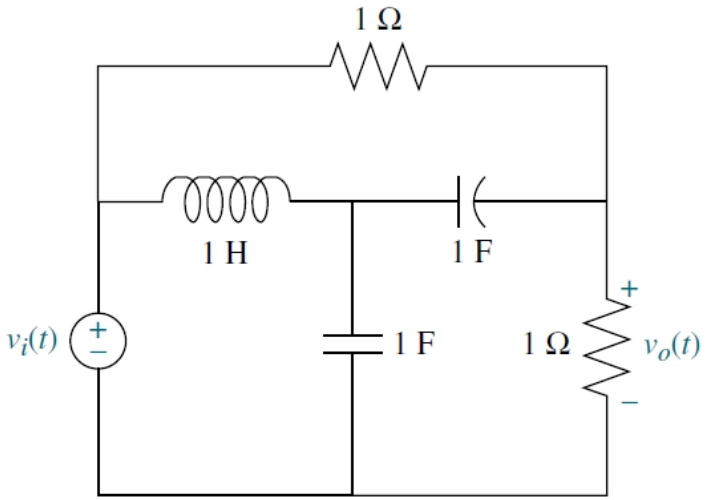
Midterm Exam
Duration: 75 minutes
(One side of an A4 sized paper is allowed as a cheat sheet)

May 3, 2023

1. (25 pts) Determine the number of roots of the following polynomial that are in the right half-plane, in the left half-plane and on the $j\omega$ axis.

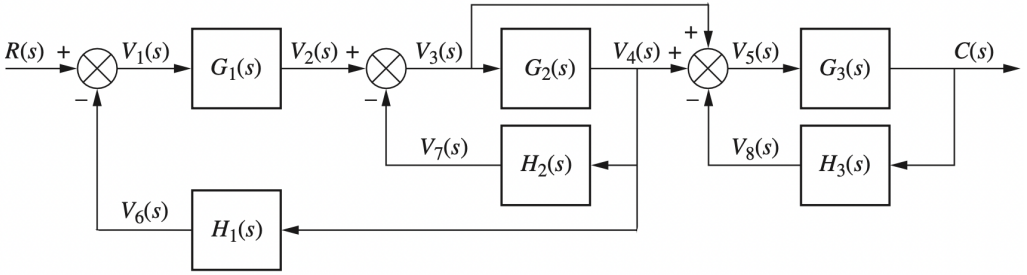
$$P(s) = s^6 - 2s^5 - 9s^4 + 6s^3 - 28s^2 + 56s + 96$$

2. (25 pts) Consider the electrical circuit given in the following figure.



Obtain the transfer function of the system where v_i is the input and v_o is the output.

3. (25 pts) Reduce the system shown in the figure below to a single transfer function.



4. (25 pts) Sketch the root locus for the unity feedback system shown in the figure below where $G(s) = \dfrac{K(s+4)}{s(s+2)}$. Find the value of K such that the unit step response critically damped.

